



NITTE
(Deemed to be University)

**NMAM INSTITUTE
OF TECHNOLOGY**

Nitte (DU) established under Section 3 of UGC Act 1956 | Accredited with 'A+' Grade by NAAC

QUESTION BANK

Subject: Basic Electronics

Course Code: EC1001-1 (3 credits)

UNIT II

1. With the help of a neat circuit diagram, derive the expression for the output voltage of non-inverting amplifier/inverting amplifier/integrator/differentiator circuit using Op-amp.
2. Design an inverting summer circuit using Op-Amp for the output voltage $V_o = -2 [0.1 V_1 + 0.5 V_2 + 2 V_3]$. Given the feedback resistor as $10 \text{ k}\Omega$. Draw the circuit diagram for the same.
3. Design an inverting adder circuit using Op-amp to obtain output voltage given by $V_o = 2V_1 + 3V_2 + 4V_3$ where V_1 , V_2 and V_3 are input voltages. Given $R_f = 20 \text{ k}\Omega$. Draw the circuit indicating all resistor values.
4. Explain the operation of non-inverting /inverting comparator with negative reference voltage with neat circuit diagram and relevant waveforms.
5. For an inverting amplifier with $R_1 = 10 \text{ k}\Omega$ and $R_f = 1 \text{ M}\Omega$, calculate the closed loop voltage gain and the required input voltage to obtain an output voltage of 3 V .
6. Find the gain and output voltage for a non-inverting amplifier using Op-amp when the input voltage is $0.7 \text{ V}_{\text{peak}}$. The supply voltage employed is $\pm 15 \text{ V}$. Given $R_f = 7 \text{ k}\Omega$ and $R_1 = 1 \text{ k}\Omega$. Sketch the waveforms.
7. An Op-Amp has a differential voltage gain of 2500 and CMRR of 30000. Calculate the common mode gain.
8. Explain the operation of non-inverting/inverting comparator with positive reference voltage with neat circuit diagram and relevant waveforms.
9. List out the ideal characteristics of an Op-Amp.
10. Briefly describe the Op-amp parameters
 - i) CMRR
 - ii) Slew Rate
 - iii) Input offset voltage
 - iv) Input offset current

11. Draw and explain the internal architecture of IC 555 timer.
12. With a neat circuit diagram explain as how IC 555 timer operates as a free running oscillator.
13. A 555 timer is configured to operate in free running mode with $R_1 = 4\text{ k}\Omega$, $R_2 = 4\text{ k}\Omega$ and $C = 0.01\mu\text{F}$. Determine the frequency and duty cycle of the output waveform.
14. For an IC 555 timer, given $D = 60\%$, $f = 2\text{ KHz}$, $R_2 = 5\text{ k}\Omega$, and $C = 0.1\mu\text{F}$.
Calculate R_1 .
15. For an IC 555 timer, $D = 75\%$, $f = 1\text{ kHz}$, $R_2 = 3.6\text{ k}\Omega$ and $C = 0.1\mu\text{F}$. Calculate T_{ON} and R_1 .
16. With an example explain the saturable property and virtual ground concept of an Op-Amp
17. Describe the operation of IC78XX series voltage regulator with feedback.
18. In a Colpitt's oscillator, $L = 2\text{ mH}$. Calculate the value of each capacitor required to generate oscillations of 1 MHz frequency. Assume $C_1 = C_2$. Write the circuit diagram.
19. With the concept of positive feedback, explain as how sustained oscillations are generated.
20. State Barkhausen's criterion for generating sustained oscillations.
21. Discuss the concept of voltage series feedback with a neat block diagram. Derive an expression for closed loop voltage gain. Comment on the closed loop gain with feedback.
22. List out the advantages of negative feedback
23. In a Colpitts oscillator, $L = 5\text{ mH}$. Find C_1 and C_2 if the frequency of oscillation is $f = 50\text{ kHz}$. Assume a feedback factor of 10% . Draw the circuit diagram for the same.
24. In a Hartley oscillator, the frequency of operation is 25 kHz . If $C = 0.02\mu\text{F}$, calculate L_1 and L_2 for 20% feedback.
25. Compute the value of R in RC phase shift oscillator for a frequency of 2 kHz and $C = 0.1\mu\text{F}$.
26. With a neat circuit diagram, explain the operation of a RC phase shift/ Hartley/ Colpitts' oscillator.